

Due by 10am Friday, 3/3

Name: _____

You may consult your class materials (notes, class packets and handouts, textbook, homework, and Moodle documents) while working on this quiz, but you may not discuss it with any one or consult any internet resources (including textbook website) outside those mentioned above. *The work you turn in must be your own.*

1. ACT, Inc. compiles data on a variety of issues in education. In 2010, the company reported that the national college first-to-second-year retention rate has historically been 77% (so 77% of first-year students stay at the same college for their second year). Consider a random sample of 400 first-year students who took the ACT exam.

(a) Describe the shape of the sampling distribution of the sample proportion of first-year students who stay at the same college for their second year. What is the mean of the sampling distribution? What is the standard deviation of the sampling distribution?

(b) What is the likelihood that your random sample of 400 first-year students would have a retention rate between 75% and 78%? Show your work and justify your answer.

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- (c) Does a college where 512 of the 603 first-year students returned the next year as sophomores have a right to brag that it has an unusually high retention rate? Justify your answer.